

THE ARENA ARMORY

AF DOSING WALL

RIVER-TOGA

PICK 2 OF 3

80 ≤60kg Cr>1.5

20 15

15-49

BEE-HELMET

5 5

2.5 2.5

80 ≤60kg Cr>1.5

EDGE-ARMOR

60 30

15-50

≤60kg

GATOR

150 110

≥80

75

15-30

VTE TREATMENT CORRIDOR

NO BRIDGE NEEDED

GREEN LANE ALL-ORAL

ORANGE LANE PARENTERAL FIRST

ROUTINE MONITORING? NO.

PROPHYLAXIS SHELF

MONITORING CORNER

AF DOSING WALL

1. Rivaroxaban (River-toga) — once daily

WHAT YOU SEE

Toga figure 'RIVER-TOGA', bottles '20', '15', '80 ≤60kg Cr>1.5'

WHAT IT ENCODES

RIVAROXABAN ('river-toga') is a direct factor Xa inhibitor dosed mostly ONCE DAILY for AF (20 mg, or 15 mg if CrCl 15–50) and TAKEN WITH FOOD to ensure absorption. It is the all-oral, single-drug option for VTE that starts with a higher loading dose then steps down. Renally adjusted; avoid in severe renal impairment.

BOARD TRAP

Rivaroxaban must be taken WITH FOOD (esp. the 15/20 mg doses) for reliable absorption — apixaban does not have this requirement. Don't forget the CrCl-based dose reduction.

2. Apixaban (Bee-helmet) — BID

WHAT YOU SEE

Bee figure 'BEE-HELMET', '5', '2.5', 'BID'

WHAT IT ENCODES

APIXABAN ('bee-helmet') is a direct factor Xa inhibitor dosed TWICE DAILY (5 mg BID; reduce to 2.5 mg BID if ≥2 of: age ≥80, weight ≤60 kg, creatinine ≥1.5). It has the LEAST renal dependence of the DOACs (~27% renal) and the most favorable bleeding profile, making it preferred in CKD and elderly patients.

BOARD TRAP

Apixaban 2.5 mg BID requires ≥2 of 3 criteria (age ≥80, ≤60 kg, Cr ≥1.5) — applying just one criterion and under-dosing risks thrombosis. Apixaban is preferred over rivaroxaban/dabigatran when renal function is poor.

3. Edoxaban (Edge-armor) — BID/QD

WHAT YOU SEE

Armored figure 'EDGE-ARMOR', '60','30', 'BID', 'P-gp inhibitors / ≤60kg / Cr<...'

WHAT IT ENCODES

EDOXABAN ('edge-armor') is a once-daily factor Xa inhibitor (60 mg, reduced to 30 mg for CrCl 15–50, weight ≤60 kg, or potent P-gp inhibitors). Notably it must NOT be used when CrCl is TOO HIGH (>95 mL/min) because reduced exposure led to higher stroke rates in AF trials.

BOARD TRAP

Edoxaban is the DOAC that is contraindicated for AF when renal function is TOO GOOD (CrCl >95) — the opposite of the usual 'reduce in renal failure' rule. This 'too clean' ceiling is a favorite board point.

4. Dabigatran (Gator) — BID

WHAT YOU SEE

Green gator figure 'GATOR', '150','110', '<150', '>80'

WHAT IT ENCODES

DABIGATRAN ('gator') is the direct THROMBIN (IIa) inhibitor, dosed 150 mg BID (110/75 mg for age/renal reduction). It is the MOST renally cleared DOAC (~80%), so it accumulates in renal failure and is the one with a specific antibody antidote (idarucizumab). Requires VTE treatment to be LED IN with parenteral heparin first.

BOARD TRAP

Dabigatran is the only listed DOAC that needs a 5–10 day parenteral (heparin/LMWH) LEAD-IN before starting (like edoxaban) — apixaban and rivaroxaban are started orally with no lead-in. Also most renal-dependent → avoid in CKD.

5. AF dosing wall — pick 2 of 3

WHAT YOU SEE

'AF DOSING WALL', 'PICK 2 OF 3', '80 ≤60kg Cr>1.5'

WHAT IT ENCODES

The AF DOSING WALL summarizes nonvalvular-AF stroke-prevention dosing and the dose-reduction triad for apixaban: pick 2 of 3 (age ≥80, weight ≤60 kg, creatinine ≥1.5) to drop to the lower dose. Each DOAC has its own renal/weight/age adjustment table — these are heavily tested.

BOARD TRAP

Reduction criteria are drug-specific — don't apply apixaban's 'pick 2 of 3' to edoxaban or dabigatran, which use CrCl thresholds and other rules.

6. No bridge needed

WHAT YOU SEE

Banner over the gate 'NO BRIDGE NEEDED'

WHAT IT ENCODES

Because DOACs have rapid onset/offset, no heparin BRIDGING is needed when starting or stopping them around procedures — you simply hold and restart. This contrasts with warfarin's slow kinetics.

BOARD TRAP

Bridging logic belongs to warfarin, not DOACs — bridging a DOAC just raises bleeding risk.

7. Parenteral-first lead-in (5–10 days)

WHAT YOU SEE

Orange lane 'ORANGE PARENTERAL FIRST', 'PARENTERAL LEAD-IN 5–10 DAYS'

WHAT IT ENCODES

VTE TREATMENT — orange (parenteral-first) lane: DABIGATRAN and EDOXABAN require an initial 5–10 day course of parenteral anticoagulation (LMWH/UFH) BEFORE the oral DOAC is started. This lead-in mirrors how those two drugs were studied.

BOARD TRAP

The lead-in DOACs are dabigatran and edoxaban. Apixaban/rivaroxaban are the all-oral, single-drug regimens (the 'green lane') — mixing these up changes the entire VTE plan.

8. All-oral green lane

WHAT YOU SEE

'GREEN LANE ALL-ORAL', River-toga & Bee-helmet figures

WHAT IT ENCODES

VTE TREATMENT — green (all-oral) lane: APIXABAN and RIVAROXABAN can be started immediately as monotherapy with NO parenteral lead-in, using a higher loading phase then a maintenance dose (apixaban 10 mg BID ×7 d then 5 mg BID; rivaroxaban 15 mg BID ×21 d then 20 mg daily).

BOARD TRAP

Don't give apixaban/rivaroxaban a heparin lead-in — they were designed as single-drug oral therapy from day 1. The loading-then-maintenance step-down is the key dosing detail.

9. VTE treatment corridor

WHAT YOU SEE

Bottom label 'VTE TREATMENT CORRIDOR', figures walking the lanes

WHAT IT ENCODES

The whole corridor frames acute VTE (DVT/PE) treatment: choose either the all-oral lane (apixaban/rivaroxaban) or the parenteral-lead-in lane (dabigatran/edoxaban), then continue anticoagulation for a defined duration (typically ≥3 months, longer/indefinite for unprovoked or ongoing risk).

BOARD TRAP

DOACs are first-line for most VTE but are NOT for severe renal failure, antiphospholipid syndrome (use warfarin), or mechanical valves (use warfarin) — know these exclusions.

10. Prophylaxis shelf (hip/knee)

WHAT YOU SEE

'PROPHYLAXIS SHELF', hip/knee bones, 'RIVA 220 QD', '10 QD', '2.5 BID', 'RIVA 5d → ...', 'ASA 81mg ×30d'

WHAT IT ENCODES

Post-orthopedic (hip/knee arthroplasty) VTE PROPHYLAXIS: rivaroxaban 10 mg once daily, apixaban 2.5 mg BID, dabigatran 220 mg once daily are all options. Duration is longer for HIP (~35 days) than KNEE (~10–14 days). Aspirin 81 mg is an accepted alternative in selected low-risk patients after an initial DOAC course.

BOARD TRAP

Prophylaxis doses are LOWER than treatment doses (e.g., apixaban 2.5 BID for prophylaxis vs 5 BID for AF) — don't confuse the two. Hip needs longer prophylaxis duration than knee.

11. Routine monitoring? NO

WHAT YOU SEE

'ROUTINE MONITORING? NO', INR/PT crossed out

WHAT IT ENCODES

A defining DOAC advantage: NO routine coagulation monitoring and no dose titration to a lab target. Fixed dosing with periodic renal/hepatic function checks is all that is required. INR/PT is NOT used to dose or monitor DOACs.

BOARD TRAP

Ordering INR to 'check' a DOAC is wrong — INR is for warfarin. A normal INR does NOT exclude a therapeutic DOAC effect.

12. PT/INR is useless for DOACs

WHAT YOU SEE

'INR/PT NO', PT(yellow)/PT(green) devices, 'CHROMOGENIC'

WHAT IT ENCODES

PT/INR and aPTT respond unpredictably to DOACs and cannot quantify their effect. If drug levels truly must be measured (e.g., before emergency surgery, suspected accumulation, extremes of weight), use a DRUG-SPECIFIC assay: anti-Xa calibrated to the specific Xa inhibitor, or dilute thrombin time / ecarin for dabigatran.

BOARD TRAP

A normal PT/aPTT does NOT rule out DOAC presence; conversely a mildly abnormal PT doesn't quantify it. Use the calibrated anti-Xa (for -xabans) or dilute thrombin time (for dabigatran) when a level is genuinely needed.

13. Anti-Xa assay (when needed)

WHAT YOU SEE

Devices labeled 'api', 'edo', 'riva', 'ANTI-Xa', 7–11h / 9–11h timing

WHAT IT ENCODES

For the factor Xa inhibitors (apixaban, rivaroxaban, edoxaban), a DRUG-CALIBRATED ANTI-Xa level is the appropriate test if measurement is required, and it should be checked relative to dose timing (peak ~1–4 h, trough before next dose). Useful in renal failure, bleeding, urgent surgery, or compliance questions.

BOARD TRAP

A generic (LMWH-calibrated) anti-Xa is not interchangeable — the assay must be calibrated to the specific DOAC. Routine levels are unnecessary; this is only for special situations.

14. Dilute thrombin time for dabigatran

WHAT YOU SEE

Devices 'dabi', 'dTT/ECT', '12h' timing

WHAT IT ENCODES

Dabigatran (anti-IIa) is best assessed, when needed, by a DILUTE THROMBIN TIME (dTT) or ECARIN clotting time, which correlate with drug concentration. A normal thrombin time (TT) effectively excludes clinically relevant dabigatran, because TT is exquisitely sensitive to thrombin inhibition.

BOARD TRAP

Use thrombin-based assays for dabigatran and anti-Xa assays for the -xabans — don't swap them. A normal standard TT rules OUT meaningful dabigatran (high sensitivity), unlike PT/aPTT.

15. Monitoring corner takeaway

WHAT YOU SEE

'MONITORING CORNER', bedside devices, INR/PT 'NO'

WHAT IT ENCODES

Bottom line locked in: DOACs are given at fixed doses with NO routine lab monitoring; check renal/liver function periodically and adjust dose by CrCl. Reserve drug-specific assays (anti-Xa for -xabans, dilute TT/ecarin for dabigatran) for emergencies, accumulation, or extremes of body weight — NEVER PT/INR.

BOARD TRAP

The exam reward is recognizing that 'no monitoring' is a benefit but NOT 'never measurable' — in the right emergency you still use the correct calibrated assay, just not PT/INR/aPTT.
